

MPhil Data Intensive Science
S2 Coursework

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1 Question 1: Accident Data

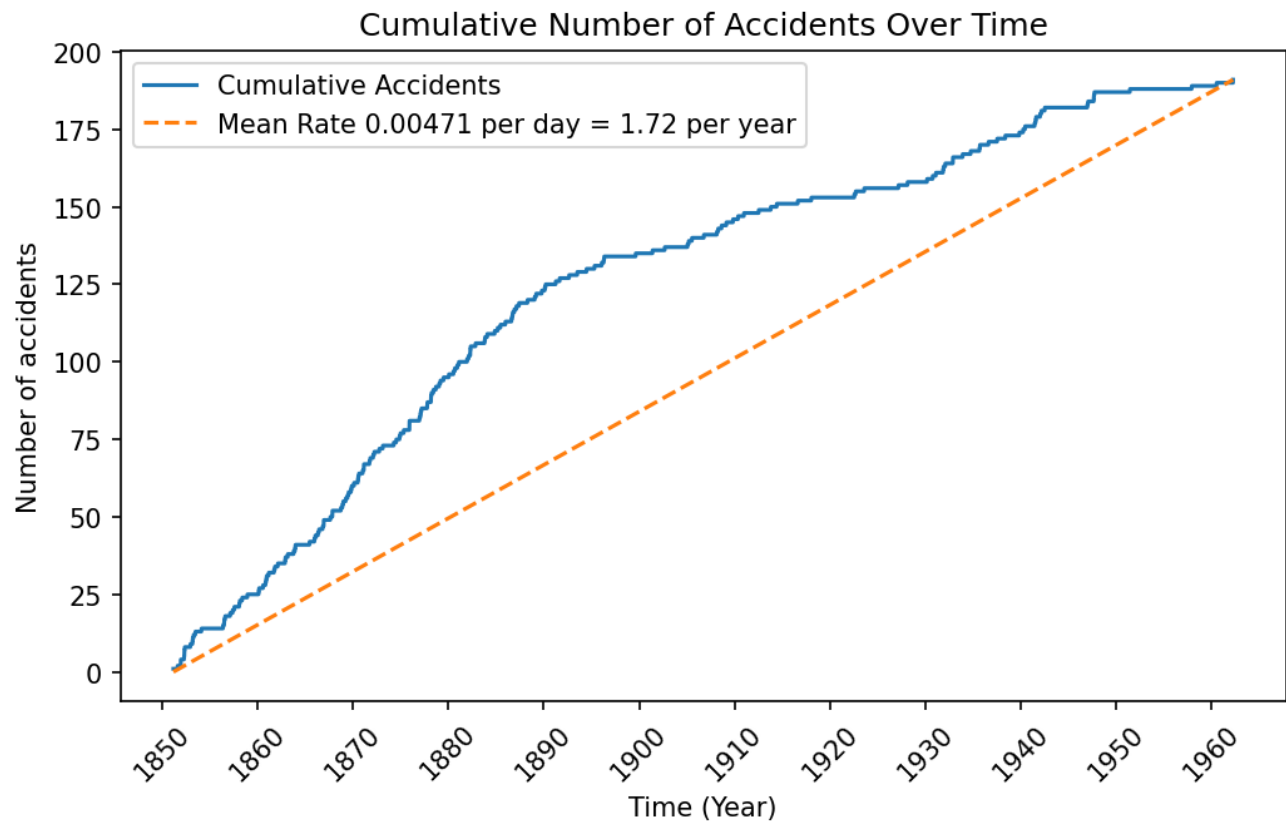


Figure 1: Q1: Cumulative number of serious mining accidents in the UK. The cumulative sum is plotted in solid blue and the mean accident rate is plotted in dashed yellow line. Deviation from the mean rate line shows that the true rate is not constant in time. The rate of accidents is higher before visibly decreasing in the 1880s.

The mean rate of accidents over the total period covered by the data is 4.71×10^3 per day or 1.72 per year.

2 Question 2: Priors

2.1 Question 2A: Even-order and Plain-order Statistics

Plain-order statistics draws k i.i.d. points from a uniform distribution and sorts them. Even-order statistics draws $2k + 1$ points, sorts them, and takes only the even-labelled points.

Even-order statistics forces drawn points to have intermediate points between them, resulting in greater separations compared to the plain-order draw as, under a uniform distribution, the probability that another point is drawn inside that small gap shrinks proportionally with the gap size. This introduces a repulsion between neighbouring change-points. Spreading out the change-points encourages the model to explore the change-point search space more thoroughly and prevents degeneracy where two change-points are clumped together such that only one of the heights affects the model, leaving the other one unconstrained by the likelihood.

When the position distributions of change-points are plotted, it can be seen that even-statistics yields more tightly distributed change-points at even spacing across our time range, yielding better coverage. This is also reflected in less extreme gap distributions: there is much less probability mass for very large or very small gaps. Furthermore, the even-statistics pulls the change-points away from the time edges, reducing the probability that the model effectively collapses to one with fewer change points and a poorly constrained height parameter.

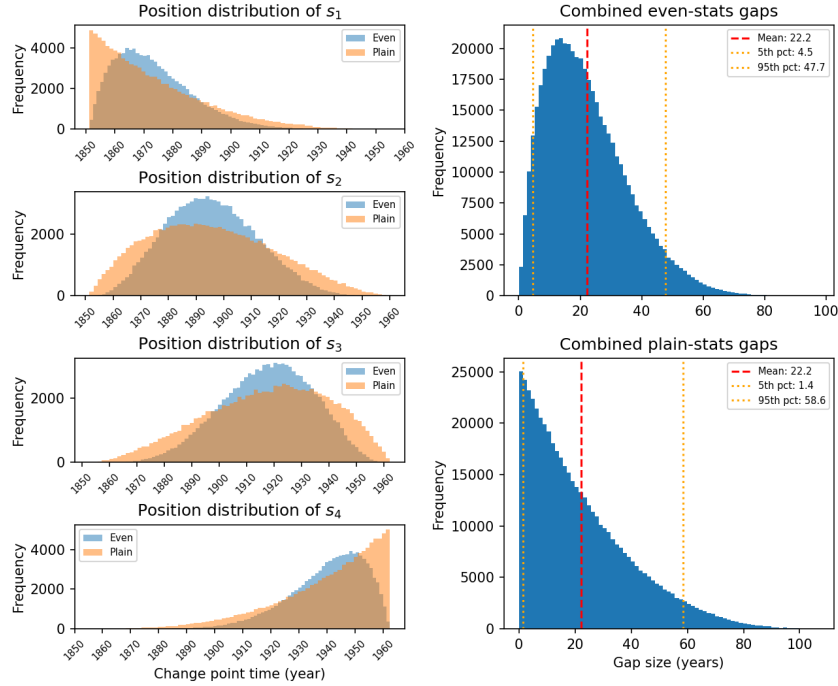


Figure 2: Q2A: Comparison of the change-points, and the implied gaps, yielded by even-order statistics and plain-order statistics priors. Right: Distributions of the positions of the change-points produced by a $k = 4$ model under even and plain-statistics with shared x -axes. We can see that the distributions of the even-statistics are more highly peaked and tighter, and that they discourage change-points from forming at the start and end extremes of our time axis. Left: The combined gaps between change-points and the time extremes. The even-stats can be seen to discourage change points that are too close together, shifting the probability mass to a more regular peaked distribution.

2.2 Question 2B: Even-order Statistics PDF Proof

The PDF of the plain-order statistics is

$$P(u_1, u_2, \dots, u_{2k+1}) = (2k + 1)!.$$

We wish to derive the PDF of even-order statistics, where only the even-indexed u are retained:

$$P(u_2, u_4, \dots, u_{2k}).$$

This can be achieved by marginalising out the variation in the odd-indexed u values. Each odd-indexed u lives between its two even-indexed neighbours, except the edge cases u_1 and u_{2k+1} which are bounded by 0 and L respectively. As such, the range of an odd-indexed point can be written as

$$u_{2j+1} \in]u_{2j}, u_{2j+2}[\quad \text{for } j \in [1, k+1]$$

where $u_0 = 0$ and $u_{2k+2} = L$.

This yields the integral to marginalise over the odd indices:

$$P(u_2, u_4, \dots, u_{2k}) = (2k+1)! \int_{u_1=0}^{u_1=u_2} \int_{u_3=u_2}^{u_3=u_4} \dots \int_{u_{2k-1}=u_{2k-2}}^{u_{2k-1}=u_{2k}} \int_{u_{2k+1}=u_{2k}}^{u_{2k+1}=1} du_1 du_3 \dots du_{2k-1} du_{2k+1}.$$

Each of the odd-indexed u are independent of each other so the integral can be factorised as

$$P(u_2, u_4, \dots, u_{2k}) = (2k+1)! \prod_{j=1}^{k+1} \int_{u_{2j-1}=u_{2j}}^{u_{2j-1}=u_{2j-2}} du_{2j-1}.$$

Because each of the factorised integrals' integrands are unity, each returns the difference between the limits

$$\int_{u_{2j-1}=u_{2j}}^{u_{2j-1}=u_{2j-2}} du_{2j-1} = u_{(2j)} - u_{(2j-2)},$$

which correspond to unscaled change-points

$$= \hat{s}_k - \hat{s}_{k-1}.$$

These can be identified as the gaps after scaling by L because each $u \in [0, 1]$:

$$L \times (\hat{s}_k - \hat{s}_{k-1}) = g_k.$$

As such, the integral is equal to the product of the gaps and since the prior is proportional to the product of the integrals, the prior is proportional to the product of the gaps:

$$\pi(s_1, s_2, \dots, s_m) \propto \prod_{k=1}^{k+1} g_k.$$

3 Question 3: The Constant Rate Model

3.1 Question 3A: Plotting Prior Vs Posterior

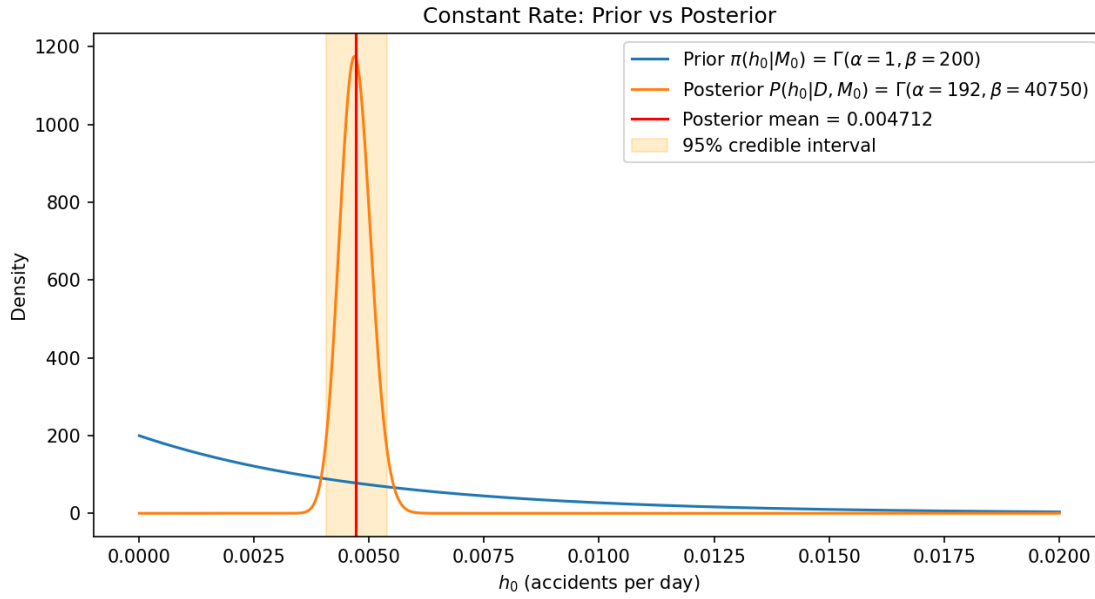


Figure 3: Q3A: Prior and posterior distributions for the height h_0 for the constant rate model. The posterior is highly peaked, with a small 95% credible interval around a mean of 0.0047. This agrees with the mean rate calculated in Q1. Since the posterior is significantly more peaked than the prior, this indicates that posterior is likelihood-dominated.

3.2 Question 3B: Calculating evidence Z_0

It is more numerically stable to deal with the log-evidence so $\ln Z_0$ was calculated instead.

$$\ln Z_0 = -1217.092$$

4 Question 4: 1 Change Point Model

4.1 Question 4A: MCMC Sampling M_1 Posterior

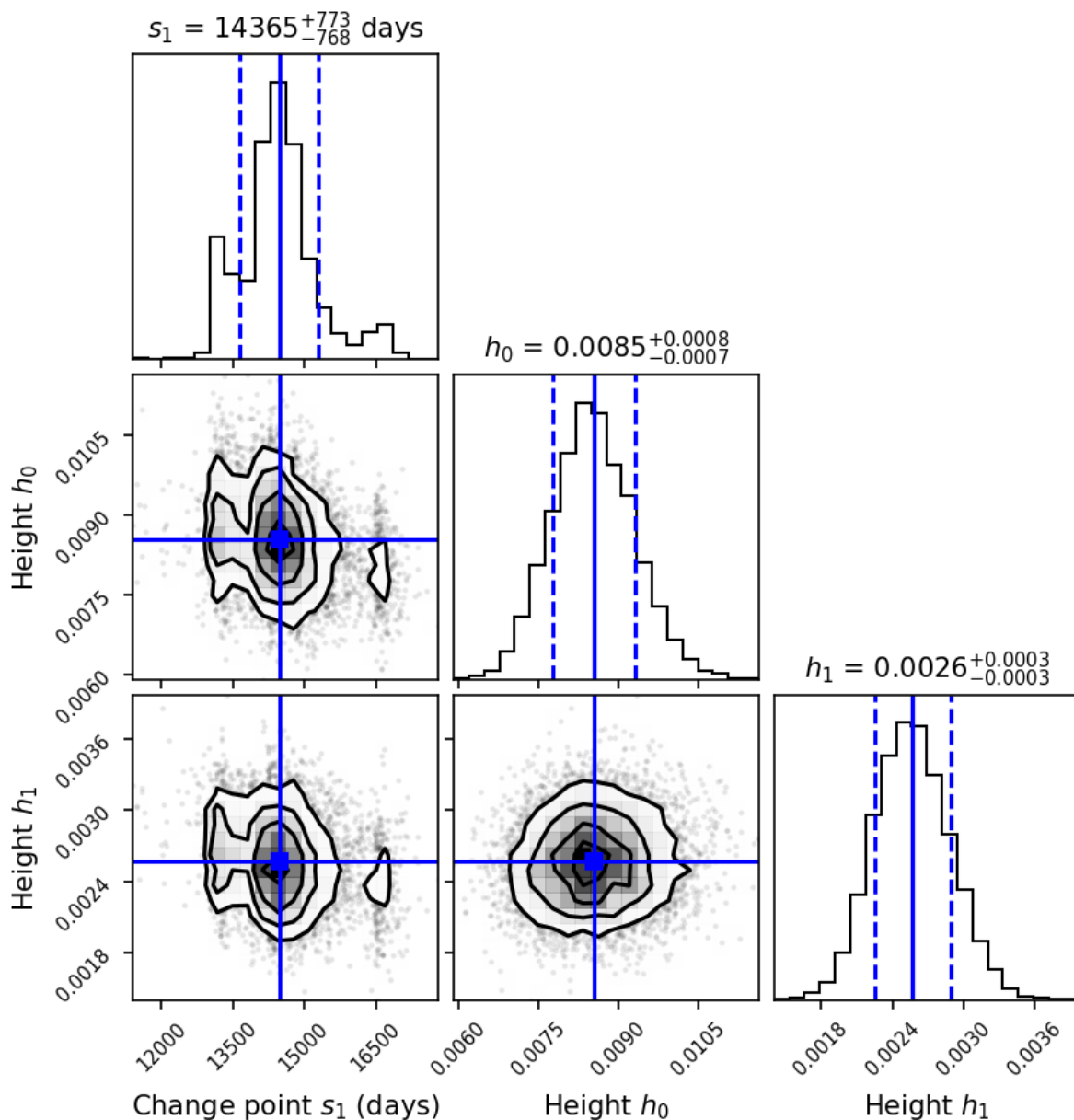


Figure 4: Q4A: MCMC-sampled posterior of the one change-point model M_1 . Each of the two heights, h_0 and h_1 , appear well-peaked but the change-point s_1 distribution is multimodal, leading to non-Gaussian joint-posterior contour plots. The dominant change-point peak agrees well with the data as it is centred around the time where the rate of accidents noticeably decreases. Correspondingly, the posterior favours $h_1 < h_0$, implying a lower accident rate after s_1 . The means are plotted in solid blue and the 1σ uncertainties are plotted with dashed lines.

The ensemble sampler from `emcee` was used. This uses an ensemble of walkers to define the proposal distribution based on the distribution of other walkers, allowing it to vary based on the scale of the posterior in each direction so we do not have to manually tune a proposal distribution to get a high acceptance ratio and fast convergence.

We only need to choose the number of walkers to balance tractability and posterior coverage. Literature suggests that $2 \times n_{\text{dimensions}}$ walkers are required at minimum. Since we are only sampling in 3D, 32 walkers,

initially distributed as the prior, was chosen to ensure good coverage. The acceptance probability achieved was above 50%, implying that enough walkers were used to yield a good picture of the posterior spread to define a good proposal distribution.

The largest autocorrelation time across the three parameters was calculated to be $\tau = 56.0$, further implying a good proposal distribution. Across 100 000 samples, this yields a respectable $N_{\text{eff}} = 1785.7$. Twice τ was chosen as a baseline for thinning to ensure independent samples. This is useful for model-comparison calculations.

A trace plot showed that the samplers converged to a tight band in all dimensions in less than 50 steps. 100 steps were discarded for safety. This trace plot also indicates good convergence. That the sampled posterior agrees with our data also indicates convergence. The mean $\bar{s}_1 = 14365$ days corresponds to a date in the 1890s, which aligns with the time when the rate of accidents in the data noticeably decreases. Similarly, the posterior favours $h_1 < h_0$, agreeing with the data that indicates a lower accident rate after the change-point.

Additional convergence diagnostics also indicate good mixing. The Geweke statistic is $G = 0.014$, close to the ideal value of 0, and the Gelman–Rubin statistics of the three dimensions are $\hat{R} = [0.999, 0.999, 0.999]$, all very close to the ideal value of 1.

4.2 Question 4B: Savage–Dickey Ratio

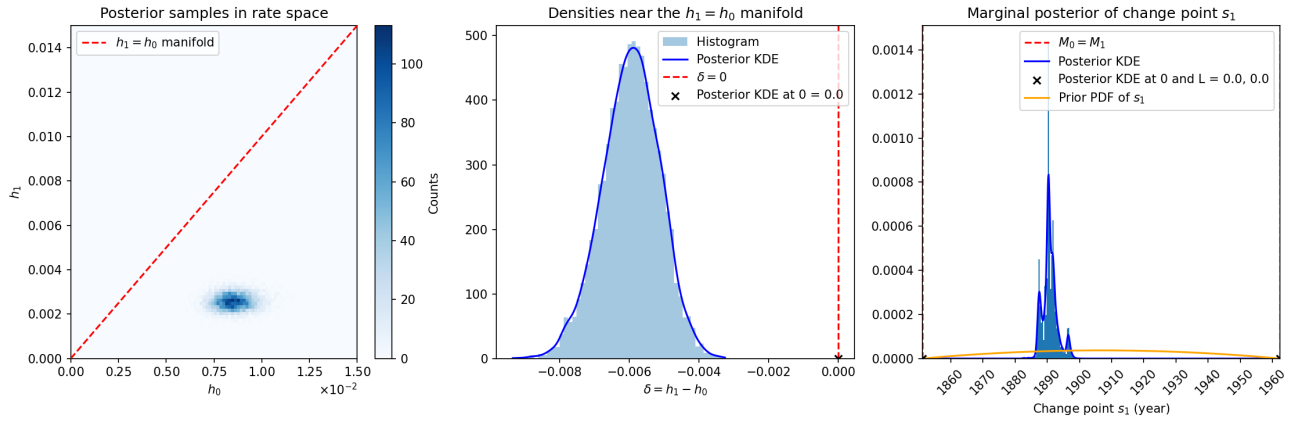


Figure 5: Q4B: Difficulty in calculating the Savage–Dickey density ratio. Left: s_1 marginalised out to yield posterior distribution in h_0 – h_1 space. We can see that all the samples fall far from the $h_0 = h_1$ manifold where $M_1 = M_0$. Middle: A plot of the distribution of $\delta = h_1 - h_0$. $M_1 = M_0$ at $\delta = 0$. We can see that the kernel density function extracted has zero-density at the target manifold. Right: The other case under which $M_1 = M_0$ is if $s_1 = 0$ or $s_1 = L$. We can see that there is no posterior or prior density in that area.

The Savage–Dickey density ratio calculates the evidence ratio between nested models, M_1 and M_0 , where model M_0 can be recovered by constraining the parameters, θ , to space ϵ .

$$M_1(\phi, \theta) = M_0(\phi) \quad \text{for } \theta \in \epsilon.$$

The ratio is given by

$$\ln Z_1 - \ln Z_0 = \ln \pi(\theta \in \epsilon \mid \phi, M_1) - \ln P(\theta \in \epsilon \mid \phi, M_1).$$

This means that posterior and prior samples in the $\theta \in \epsilon$ regions are needed to estimate the density ratio. M_1 reduces to M_0 when $h_0 = h_1$, or when $s_1 = 0, L$. As such, the posterior and prior must be sampled on these manifolds.

However, because the prior is proportional to the product of the gaps, the prior for M_1 is $\propto s_1 \times (L - s_1)$. This is 0 at $s_1 = 0, L \in \epsilon$ and cannot be sampled. Similarly, computing a KDE on the marginalised sampled posterior for s_1 yields 0 probability density there. Examining the rate-space marginal posterior shows that all samples lie far from the $h_1 = h_0$ manifold. Defining $\delta = h_1 - h_0$ and computing a KDE over the posterior samples also yields 0 probability density at $\delta = 0 \in \epsilon$.

This means that it is impossible to sample the posterior and prior at $\theta \in \epsilon$, making it impossible to calculate the Savage–Dickey density ratio between M_1 and M_0 .

4.3 Question 4C: Laplace Approximation for Evidence Calculation

The Laplace approximation works by approximating the posterior peak as a multivariate Gaussian distribution and using the tractable Gaussian integration to approximate the evidence. However, the sampled posterior here is clearly non-Gaussian. The s_1 distribution is multimodal and skewed, producing irregular contours. The piecewise form of $x(t)$ in the likelihood function also complicates the curvature of the posterior. As such, the volume of the posterior cannot be cleanly approximated by a single multivariate Gaussian.

The Laplace approximation also does not account for the physical parameter constraints. h_0 and h_1 are rate parameters and must be greater than 0 while s_1 is constrained between 0 and L , yet the multivariate Gaussian the Laplace approximation attempts to use has support everywhere. The heights are both centred very close to 0, so any Gaussian approximation centred there will have non-trivial probability density outside the physical constraints.

4.4 Question 4D: Nested Sampling

Algorithm Details

Nested sampling computes the evidence by transforming the many-dimensional parameter integral into 1D over the prior mass. This integral is approximated by drawing live points: at each iteration, the lowest likelihood point is removed and a new one is drawn from a region of higher likelihood. By tracking the likelihoods of these removed points, the evidence can be calculated by numerical integrating them over analytic estimates of the prior mass change.

Dynesty was chosen to perform the nested sampling. It requires the log-likelihood and a prior PPF. The heights are Γ -distributed, which has an analytic PPF. However, the change-points are Dirichlet-distributed, which doesn't have a simple analytic PPF. However, because M_1 has only one change-point, the Dirichlet distribution reduces to a Beta distribution, and the analytic PPF for the Beta distribution can be used.

Dynesty samples efficiently from the likelihood-constrained region by defining a set of ellipsoids bounding the live points and drawing from within that region. The 'multi' ellipsoid bound was used because the distribution is multimodal and cannot be approximated by just one ellipsoid. Because the data are low-dimensional, efficient uniform sampling within the bounds is practical. 5000 live points were used, resulting in a very low uncertainty $\ln Z_1 = -1187.31 \pm 0.13$, indicates that the nested sampling run has stably estimated $\ln Z_1$.

Nested Sampling Results (excluded from word count but included for completeness)

The prior over the models is given by

$$\pi(k) \propto \frac{\lambda^k e^{-\lambda}}{k!},$$

where $\lambda = 3$.

Analytically, the prior ratio between M_1 , $\pi(k = 1)$, and M_0 , $\pi(k = 0)$, is

$$\frac{\pi(k = 1)}{\pi(k = 0)} = \frac{\lambda^1 e^{-\lambda}/1!}{\lambda^0 e^{-\lambda}/0!} = \frac{\lambda e^{-\lambda}}{e^{-\lambda}} = \lambda = 3.$$

As such, the posterior odds ratio is three times the evidence ratio:

$$\frac{P(M_1 | \text{data})}{P(M_0 | \text{data})} = \frac{Z_1}{Z_0} \cdot \frac{\pi(k = 1)}{\pi(k = 0)} = \frac{Z_1}{Z_0} \cdot \lambda = 3 \cdot \frac{Z_1}{Z_0}.$$

Using nested sampling, the evidence for the one-change-point model was found to be

$$\ln Z_1 = -1187.34 \pm 0.30,$$

while for the zero-change-point model

$$\ln Z_0 = -1217.09.$$

This gives a log evidence ratio of

$$\ln\left(\frac{Z_1}{Z_0}\right) = 29.75 \pm 0.30,$$

corresponding to an evidence ratio of

$$\frac{Z_1}{Z_0} = (8.35 \pm 2.50) \times 10^{12}.$$

Including the prior ratio, the log posterior odds ratio is

$$\ln\left(\frac{P(M_1 \mid \text{data})}{P(M_0 \mid \text{data})}\right) = 30.85 \pm 0.30,$$

which corresponds to a posterior odds ratio of

$$\frac{P(M_1 \mid \text{data})}{P(M_0 \mid \text{data})} = (2.51 \pm 0.75) \times 10^{13}.$$

5 Question 5: RJMCMC

5.1 Question 5B: MCMC Trace Plot

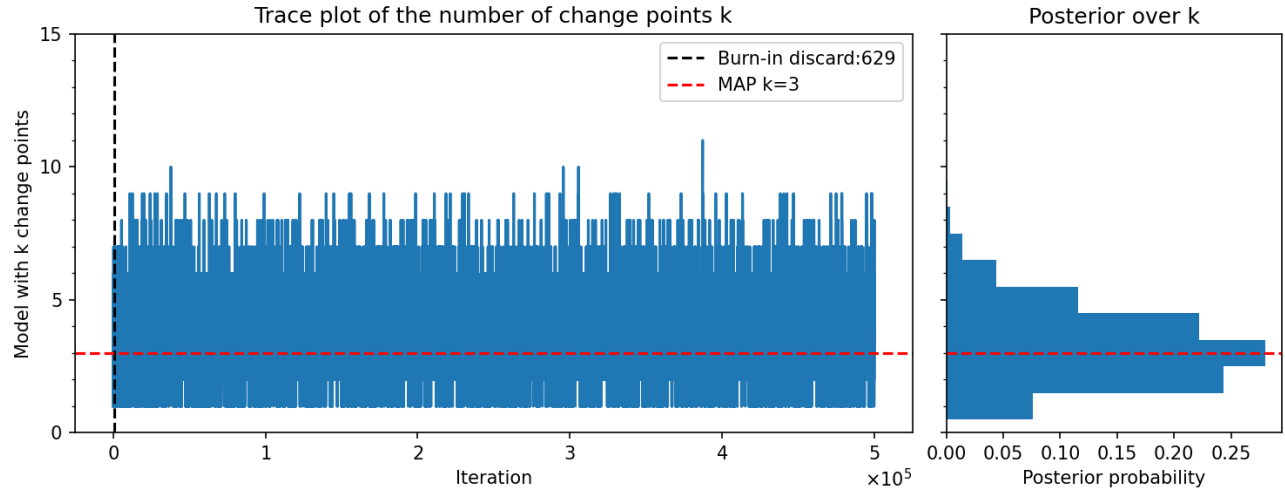


Figure 6: Q5B: Left: Trace plot showing the marginal trajectory of the walker through k . Right: The number of samples in each k plotted to share the same y -axis. The MAP at $k = 3$ is indicated with a dashed red line.

5.2 Question 5C: Inferred Rate

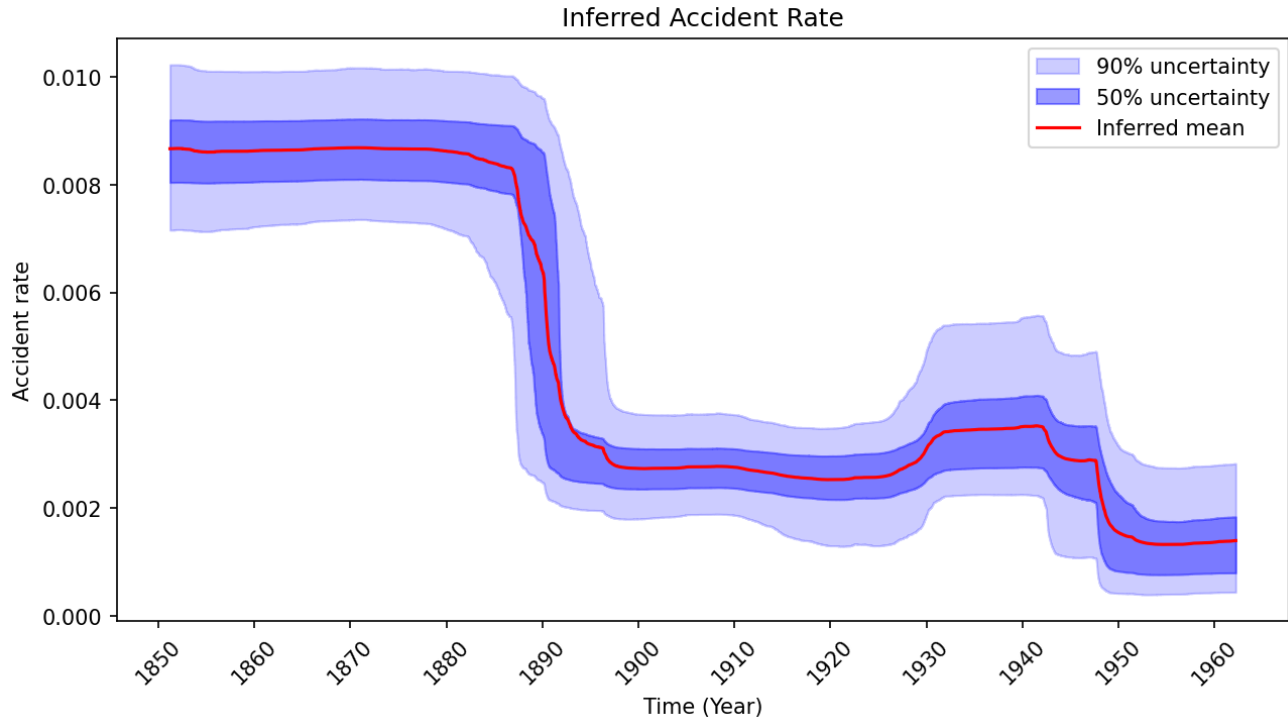


Figure 7: Q5C: The inferred mean accident rate, plotted in red, and its 50% and 90% uncertainty bands. The inferred accident rate drops sharply in the 1890s, agreeing with the data.

The posterior predicted accident rate is calculated by marginalising the posterior samples over k . For tractability, the autocorrelation time τ was calculated and the chain was thinned by 2τ . The first 10τ samples were discarded to ensure that the burn-in was removed.

There is now a set of retained samples

$$\{x^{(m)}\}_{m=1}^M \quad \text{where } x^{(m)} \in \mathbb{R}^L.$$

Each is a piecewise function returning

$$x^{(m)}(t) = h_j^{(m)}, \quad s_{j-1}^{(m)} \leq t < s_j^{(m)},$$

with $s_0^{(m)} = 0$ and $s_{k+1}^{(m)} = L$.

The inferred rate at time t is taken to be the mean across sampled models:

$$\bar{x}(t) = \frac{1}{M} \sum_{m=1}^M x^{(m)}(t).$$

The uncertainty regions are also calculated from the samples by taking the percentiles for sets in slices of t :

$$\{x^{(m)}(t)\}_{m=1}^M.$$

The 5th and 95th percentiles were taken for the 90% uncertainty, and the 25th and 75th percentiles were taken for the 50% uncertainty. These variations in the model's x values reflect the intra-model uncertainties in height and change point position of the sampled models, yielding an uncertainty in the overall inferred value.

6 LLM Usage Declaration

LLMs, specifically Claude and GPT, were used while doing this coursework. They were used for identifying and suggesting fixes for particularly esoteric bugs I encountered, helping me with $\text{\LaTeX}$, writing nice docstrings, and implementing some of the more avant-garde `matplotlib` functions. The derivations, thinking, and analysis are my own.